Ans. (d) : From S_1 -

$$\left.\begin{matrix} C < D \\ C > A \\ C > B \end{matrix}\right\}$$

From, S_2 - D is eldest of all,

S_3 - A > B,

S_1 & S_3

$$\boxed{B < A < C < D}$$

So, S_1 & S_3 together are sufficient to answer the question.

35. Two Statements S_1 and S_2 are given below with regard to four numbers P, Q, R and S followed by a Question :

S_1 : R is greater than P as well as Q.

S_2 : S is not the largest one.

Question:

Among four numbers P, Q, R and S, which one is the largest?

Which one of the following is correct in respect of the above Statements and the Question?

(a) S_1 alone is sufficient to answer the Question.

(b) S_2 alone is sufficient to answer the Question.

(c) S_1 and S_2 together are sufficient to answer the Question, but neither S_1 alone nor S_2 alone is sufficient to answer the Question.

(d) S_1 and S_2 together are not sufficient to answer the Question.

Ans. (c) : From S_1

$$\text{and } \left.\begin{matrix} R > P \\ R > Q \end{matrix}\right\}$$

From S_2

S is not the eldest

$\therefore$ R will be the eldest.

Hence, S_1 & S_2 together are sufficient to answer the question. But neither S_1 alone nor S_2 alone is sufficient to answer the question.

36. Two Statements S_1 and S_2 are given below followed by a Question :

S_1 : n is a prime number.

S_2 : n leaves a remainder of 1 when divided by 4.

Question:

If n is a unique natural number between 10 and 20, then what is n?

Which one of the following is correct in respect of the above Statements and the Question?

(a) S_1 alone is sufficient to answer the Question.

(b) S_2 alone is sufficient to answer the Question.

(c) S_1 and S_2 together are sufficient to answer the Question, but neither S_1 alone nor S_2 alone is sufficient to answer the Question.

(d) S_1 and S_2 together are not sufficient to answer the Question.

Ans. (d) : It's given that n is a unique natural number between 10 & 20. According to S_1 n is a prime number So, our number n may be : 11, 13, 17 on 19 According to S_2, on dividing n by 4 we get a remainder of 1.

So, n can be : 13, 17

Either of the two statement alone is not sufficient to answer the question. Even if we read both the statements together, we do not get a unique value of n. Instead, we end up with two possible values of n, i.e. 13 & 17. Hence, we can say that even S_1 & S_2 together are not sufficient to answer the question. Hence, option (d) is the answer.

37. In the sum

$\otimes + 1\otimes + 5\otimes + \otimes\otimes + \otimes 1 = 1\otimes\otimes$

for which digit does the symbol $\otimes$ stand?

(a) 2 (b) 3

(c) 4 (d) 5

Ans. (b) : Suppose that $\otimes = x$

Then $x + (10 + x) + (50 + x) + (10x + x) + (10x + 1)$

$= 100 + 10x + x$

$= 24x + 61 = 100 + 11x$

$= 13x = 39$

$x = \frac{39}{13} = 3$

$\otimes = 3$

38. Two Statements S_1 and S_2 are given below followed by a Question :

S_1 : There are not more than two figures on any page of a 51-page book.

S_2 : There is at least one figure on every page.

Question:

Are there more than 100 figures in that book?

Which one of the following is correct in respect of the above Statements and the Question?

(a) Both S_1 and S_2 are sufficient to answer the Question, but neither S_1 alone nor S_2 alone is sufficient to answer the Question.

(b) S_1 alone the sufficient to answer the Question.

(c) S_1 and S_2 together are not sufficient to answer the Question.

(d) S_2 alone is sufficient to answer the Question.

Ans. (c) : As per statement 1, there are 2, 1 or 0 figures on each page. There are 51 pages in the book.

So, maximum possible images in the book

$51 \times 2 = 102$

& the minimum possible images in the book = 0

So, statement 1 alone is not sufficient. As per statement 2, there is 1, 2, 3, 4…….∞ images on each page. But we do not know the number of pages in the book.

39. A family of two generations consisting of six members P, Q, R, S, T and U has three males and three females. There are two married couples and two unmarried siblings. U is P's daughter And Q is R's mother-in-law. T is an unmarried male and S is male. Which one of the following is correct?

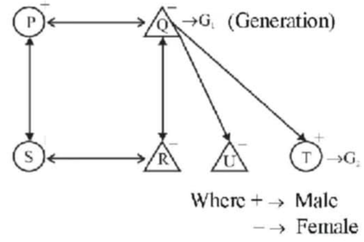
(a) R is U's husband.

(b) R is S's wife

(c) S is unmarried.

(d) None of the above

Ans. (b) :



Hence, R is S's wife.

40. A car travels from a place X to place Y at an average speed of v km/hr, from Y to X at an average speed of 2v km/hr, again from X to Y at an average speed of 3 v km/hr and again from Y to X at an average speed of 4v km/hr. then the average speed of the car for the entire journey.

(a) is less than v km/hr/ v km

(b) lies between v and 2v km/hr

(c) lies between 2v and 3v km/hr

(d) lies between 3v and 4v km/hr

Ans. (b) : Let the distance between place X and place Y = d km

$\because$ Average speed $= \frac{\text{Total Distance}}{\text{Total Time}}$

According to question,

$$\text{Average speed} = \frac{d+d+d+d}{\frac{d}{v}+\frac{d}{2v}+\frac{d}{3v}+\frac{d}{4v}}$$

$$\frac{4d}{\frac{d}{v}\left[\frac{1}{1}+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}\right]}$$

$$\frac{4v}{\left(\frac{12+6+4+3}{12}\right)}$$

$$\Rightarrow \frac{4}{\frac{25}{12}}$$

$$\Rightarrow \frac{4v \times 12}{25}$$

$$= \frac{48v}{25}$$

$= 1.92$ V km/hr

Hence, The average speed of the car for the entire journey lies between v and 2v km/hr.

41. Consider the following statements:

1. The minimum number of points of intersection of a square and a circle is 2.

2. The maximum number of points of intersection of a square and a circle is 8.

Which of the above statements is/are correct?

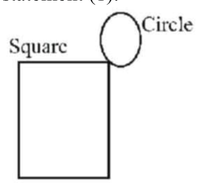
(a) 1 only

(b) 2 only

(c) Both 1 and 2

(d) Neither 1 nor 2

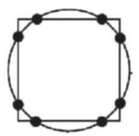
Ans. (b) : From statement (1):-



The minimum number of points of the intersection of a square and a circle is one.

Hence statement 1 is not correct.

From statement (2):-



"The maximum number of points of intersection of a square and a circle is 8.

Hence, Both 1 and 2 are correct.

42. How many different 5-letter words (with or without meaning) can be constructed using all the letters of the word 'DELHI' so that each word has to start with D and end with I?

(a) 24 (b) 18

(c) 12 (d) 6

Ans. (d) : The word "D E L H I"

According to question,

each word has to start with 'D' and end with 'I'.

$\therefore$ D _ _ _ I

Then,

The number of different 5 - letter words (with or without) meaning = 31

$= 3 \times 2 \times 1 = 6$